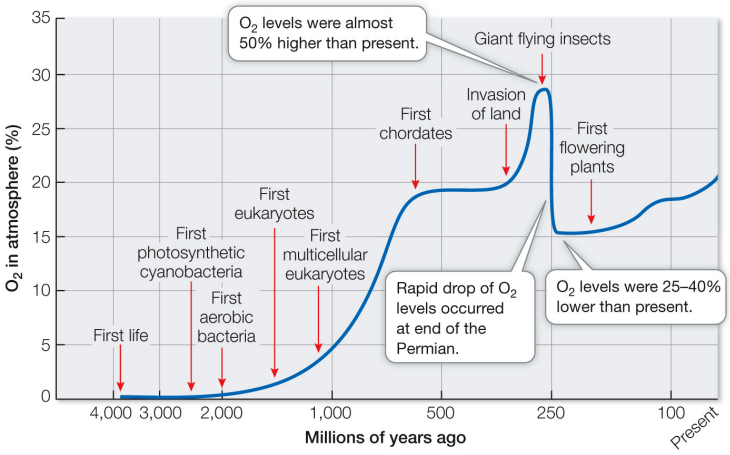
Study Guide for The Anthropocene and Conservation

This study guide provides examples for the topics that you might see on the final exam over the Anthropocene and Conservation lectures. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

1. Put the events below in the order in which they occurred in Earth's history.
    a. The origin of life on Earth
    b. The evolution of photosynthesis
    c. The colonization of land by early tetrapods
    d. The development of fossil fuels (oil, etc)
    e. Mass extinction

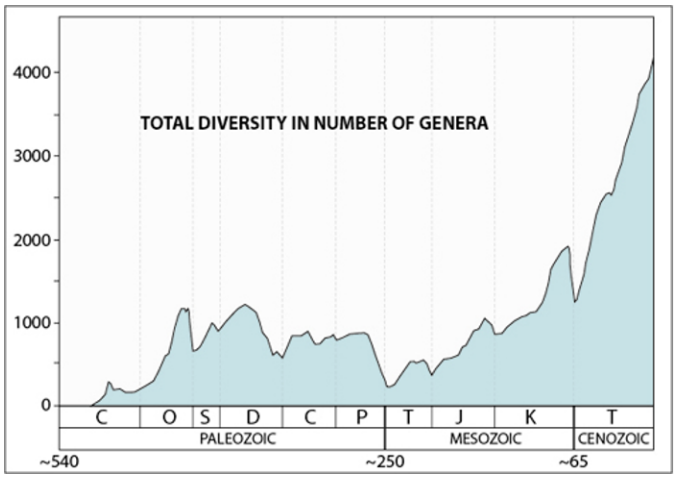
2. The diagram below shows the O_2 levels in the atmosphere throughout Earth's history. Important biological events are labeled.



    a. Photosynthesis evolves after the first evidence of life. If they were not photosynthetic, how did the first organisms obtain sugar for respiration?
    b. The first evidence of aerobic respiration occurs about 2 billion years ago. Why did aerobic respiration arise after photosynthesis? If the first living organisms were not aerobic, how were they able to perform respiration?
    c. Why do we see an increase in O2 levels following the invasion of land? What was the terrestrial landscape like at this time?
    d. Why do O2 levels drop so significantly ~250 mya? Why do we not see similar drops 444 mya, 359 mya, 201 mya, or 65 mya?

3. The graph below shows the total number of genera (y axis) on Earth from ~540 million years ago to today (x axis).
    a. The largest increase in genera is seen in the recent past, since the end of the Cretaceous Period. Why then is the Cambrian Period considered an 'explosion' of life?

    b. What is the overall pattern in the figure? What does this indicate about the resilience of life on Earth?



4. Describe how the diversification of angiosperms in the Paleogene influenced the terrestrial environment and the diversification of other taxonomic groups.

5. What is the Anthropocene?

6. Give an example of each of the following responses to environmental change driven by human activity
    a. Move
    b. Acclimate
    c. Adapt
    d. Die

7. Compare and contrast the California Species of Special Concern, Endangered Species List, CITES-listed species, and the IUCN Red List.

8. What is the extinction vortex? What happens to a population that is in the vortex?

9. Describe an example of how each of the factors below threatens biodiversity
    e. Habitat loss and alteration
    f. Climate change

10. What is the primary goal of California's 30x30 initiative? Are we on track to meet this goal?

11. Fragmentation of habitat is a major problem for conservation. What strategies have been used to reduce fragmentation of remaining habitat? Phrased another way, how do conservation land managers increase connectivity between habitats?